

TABIA TANZIN PRAMA

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RESEARCH INTEREST

Natural Language Processing, Computational Social System, Human-computer interaction, Explainable AI and Medical Informatics.

EDUCATION

- Jahangirnagar University** Dhaka, Bangladesh
• *M.Sc in Computer Science and Engineering; CGPA:3.88(Rank: 2/65)* 2022 -2024
Thesis Dissertation: HEALTH.AI: Response-Act Guided Adolescent Health Counseling System Using LLM Model. Developed a novel Response-Act guided Reinforced Dialogue Generation model for adolescent health counseling, utilizing transformer architecture and Proximal Policy Optimization.
- Jahangirnagar University** Dhaka, Bangladesh
• *B.Sc in Computer Science and Engineering; CGPA: 3.70 (Rank: 5/65)* 2018 - 2021
Thesis Dissertation: Towards Explainable AI in Identifying and Quantifying the Degree of Cyberbullying.

SELECTED PUBLICATIONS

- **AI-Enabled Deep Depression Detection and Evaluation Informed by DSM-5:** Tabia Tanzin Prama, Md. Saiful Islam, Senior Member, IEEE, Md Musfique Anwar and Ifrat Jahan, IEEE Transactions on Computational Social Systems. [\[Link\]](#)
- **Sylheti to Standard Bangla Neural Machine Translation: A Deep Learning-Based Dialect Conversion Approach.:** Tabia Tanzin Prama and Md Musfique Anwar. Hybrid Intelligent Systems - 23rd International Conference on Hybrid Intelligent Systems (HIS 2023). [\[Link\]](#)
- **SobdoKrom: An Unsupervised Bengali Keyword Extraction Model using Pre-trained Large Language Model:** Tabia Tanzin Prama and Md Musfique Anwar. Hybrid Intelligent Systems - 23rd International Conference on Hybrid Intelligent Systems (HIS 2023). Rank-C [\[Accepted\]](#)
- **Deep Learning-Based Classification of Conference Paper Reviews: Accept or Reject?:** Tabia Tanzin Prama, Al Amin Biswas and Md Musfique Anwar. Intelligent Systems Design and Applications - 23rd International Conference on Intelligent Systems Design and Applications (ISDA 2023). Rank-C [\[Accepted\]](#)
- **An AI-Based Clinical Recommendation System Using Ensemble-Based Soft Voting Classifier:** Tabia Tanzin Prama, Farhana Sarkar, Khondaker A. Mamun. Trends in Electronics and Health Informatics 2023. [\[Accepted\]](#)
- **Evaluation of Machine Learning Algorithms for Early Prediction of Type-2 Diabetics with Associated Factors.:** Tabia Tanzin Prama, Farhana Sarkar, Khondaker A. Mamun. IDF Virtual Congress 2023 on Diabetes Complications and Diabetes in Crisis, December 4-7, 2023. Rank-C [\[Best Poster Paper\]](#)
- **Modeling Information Credibility in Time-Specific Online Social Media based on Structural and Attribute Properties:** Md. Habibur Rahman, Tabia Tanzin Prama and Md Musfique Anwar. International Journal of Computer Information Systems and Industrial Management Applications, 2021. Rank-Q3 [\[Link\]](#)
- **An Integrated Expert System with a Supervised Machine Learning-based Probabilistic Approach to Play Tic Tac Toe:** Muhammad Sakib Khan Inan; Rizwan Hasan; Tabia Tanzin Prama. 2021 IEEE 12th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, USA, December 2021 [\[Link\]](#)
- **Modeling Topic Specific Credibility in Twitter based on Structural and Attribute Properties:** Md. Habibur Rahman, Tabia Tanzin Prama and Md Musfique Anwar. 20th International Conference on Hybrid Intelligent Systems (HIS 2020) 2020, December 14-16, 2020 [\[Link\]](#)
- **Modeling Trustworthiness of Topics Based on Structural Properties in Twitter(oral presentation):** Md. Habibur Rahman, Tabia Tanzin Prama and Md Musfique Anwar. IEEE International Symposium on Technology and Society (IEEE ISTAS20)2020, 12-15 November 2020, Arizona, USA [\[Link\]](#)

PROJECTS

- **E-commerce Web (Django, HTML, CSS, JS, Python, Bootstrap)** 🐛: Team-27 design and develop a **dynamic** online e-commerce website with live Chatting, Blogging, Payment Method, and Shipping Procedure.
- **WeCare Mobile Application (Deep Learning, JAVA)** 🐛: Develop **breast cancer** prevention app employing DL model to cancer with **96% accuracy**, suggestions for Nearest hospital suggestions using **MapBox** and women community.
- **Easy E-learning Mobile application (JAVA, API)** 🐛: Develop a help aid app for visually impaired students with speech of pdf text conversion, friendly font generation, Audio recording or text to speech feature using **Google API**
- **Covid-19 Info Chat Bot (chat-fuel)** 🐛: Develop a **dynamic** peer to peer data sharing platform to serve COVID-19 related information, ambulance service, health service in Bangladesh

TEACHING EXPERIENCE

- **International University of Business Agriculture and Technology (IUBAT)** Dhaka, Bangladesh
• *Lecturer - Computer Science and Engineering;* Nov 2023 - Present
 - Conduct courses on **Artificial Intelligence, Operating System, Data structure and Algorithm**, supervise final year projects and administrative tasks like routine, course outline preparation

RESEARCH EXPERIENCE

- **Research Engineer, AIMS Lab, UIU** May 2023 - Present
 - Build a **multilingual AI engine** using **LLM model** to address adolescent Sexual and Reproductive Health and Mental Health challenges(Funded by the **Bill & Melinda Gates Foundation**)
 - Develop a digital platform with an Ensemble Model for predicting **treatment option** using Demographics and Symptoms, a **medicine tracker**, and **telemedicine service** to reduce Antimicrobial resistance.
 - Proposed diabetes prediction model with associated factors using **XAL and machine learning approach**.
- **Research Assistant, JU Data Mining Lab** July 2021 - Present
 - Designed a unique bidirectional conversion system for four Bangladeshi dialects, employing **text-to-text** with **word-to-word mapping**, backed by a **bilingual dictionary** and rule-based **morphological transformations**
 - Designed an unsupervised model to extract keywords from Bangla content using LLM model.
 - Proposed an **XAL based deep learning** approach to classify peer review of conference paper.
 - Design questionnaire, explore significant factors related to course and teacher using **machine learning** and **XAI** tools.
- **Research Assistant, The University of Newcastle, Australia** July 2021 - 2023 Feb
 - We introduce a novel framework based on **emotional features, topical events, and behavioral-biometric signals** to measure intensity of depression under the supervision of **Dr Saiful Islam**
 - We incorporate (**DSM-5-TR**)clinical assessment procedure for automatic depression screening and monitoring.
- **Research Assistant, BCS2 Research Lab** July 2021 - 2023 Feb
 - Analyzing AI ethics papers and ethical principles of firms using **HDP-Topic Modeling, Embedding Regression, and Readability Analysis** to detect ethics washing in AI research under the supervision of **Dr Nur Ahmed**.

HONORS AND AWARDS

- **ICT Research Fellowship:** Nomination by ICT ministry, Bangladesh for Research Fellowship in ICT. March, 2024.
- **IDF 2023:** Awarded best poster paper award in IDF World Diabetes Congress 2023.-Dec,2023
- **INNOVATION CAMP'20: Champion** at GIRLS INNOVATIVE BOOT CAMP-1 organised by BDOSN.
- **NST Fellowship:** Awarded a **research** fellowship from **Ministry of Science and Technology, Bangladesh**
- **ICC'20: Best Idea Award** at Idea Contest (Top 10), INTERNATIONAL COVID-19 CONGRESS BY IEEE.
- **Ada Lovelace'21:** Ranked **6th** place in Ada Lovelace Datathon 2021 hosted by **HerWill**.
- **CSE FEST'20: First Runner's Up** at NATIONAL GIRLS CSE FEST SCIENTIFIC POSTER PRESENTATION
- **SCIENTIFIC POSTER PRESENTATION:** First Runner's Up at **MUJIB CARNIVAL**
- **GWOC'21:** Ranked top 10% out of 8000+ participants in **Machine learning & Data Science** categories.
- **NGPC'19:** Ranked in top 10% out of 105+ teams in National Girls Programming Contest 2019.
- **Messaging'20:** Participated at the Facebook Hackathon: Messaging 2020
- **Merit Scholarships:** Government Board Merit scholarships for outstanding results in both Higher Secondary Certificate (2017) and Secondary School Certificate (2015) .

SKILLS SUMMARY

- **Languages:** Python, Java, C, C++, Matlab, SQL, HTML
- **Machine Learning:** Scikit, NLTK, SpaCy, TensorFlow, Keras, Django,PyTorch, Keras, OpenCV
- **Developer Tools:** VS Code, Android Studio, LATEX, PyCharm, Jupyter Notebook, Spyder, Weka, Eclipse
- **Version Control:** Git, Github

VOLUNTEER EXPERIENCE

- **Bangladesh Youth Leadership Training Program** The University of New York
Received a 3-month long youth Leadership Training Program on technical & soft skills training Oct 2019 – Jan 2020
- **JU Computer Club** Jahangirnagar University
As Assistant Secretary, Conducted online and offline technical & soft-skills training program June 2018 – Present
- **Event Organizer at JU CSE Fest-2019** Dhaka, Bangladesh
Organized events and conducted Project Showcasing competition March 2019